

Adam Nelson

📍 Hamilton, Ontario, Canada ✉ nels123159@gmail.com ☎ (905)921-7809 🌐 in/adam-mark-nelson/ 📄 https://amn6.github.io/

SUMMARY

Senior Software Engineer with 4 years of experience in developing firmware for AMD products. Skilled in embedded software development, debugging, hardware testing, and root cause analysis. Additionally developed android applications to help verify hardware accelerated processes on AMD chips. Proven track record of developing firmware with high-performance and reliability. Experienced in working with multiple teams to ensure the successful completion of projects. Quick to adapt to new roles, and self-driven.

EXPERIENCE

Senior Software Engineer

AMD

July 2023 - Present, Markham, ON

- Designed test cases for used by all display devices (GPU/APU) designed at AMD.
- High knowledge of Display Port 2.1 and HDMI 2.1 standards.
- Verified new hardware from pre-silicon design stage, to post-silicon stage with OS(Linux) drivers and firmware; with tests written in C/C++.
- Implemented and maintained Python and Bash scripts automating test suite decreasing overall time by 50%.
- Aided development on an AI image classifier to detect corrupt display output.

Firmware Engineer

AMD

May 2021 - June 2023, Markham, ON

- Developed the Firmware for Video Encoding and Decoding on the next generation consoles (PS5, XBOX ONE).
- Familiar with AVC, HEVC, and VP9 codecs.
- Ran and organized the weekly sync-up meeting for display diagnostics team.

Software Engineer Intern

AMD

May 2019 - August 2020, Markham, ON

- App Development, last task of the internship involved creating an Android Application in Java to test video Encoding/Decoding, and 3D Rendering.
- Automated performance tests, increasing productivity by increasing frequency from weekly to each commit.
- Created CI/CD with Jenkins and Python scripts. Improved the speed of software development process by 40%, reduced time to deploy features from 1 week to 2 days.

PROJECTS

Strain Characterization Apparatus

McMaster University Capstone • https://www.youtube.com/watch?v=o3wQCIq5QQM&ab_channel=AdamNelson • September 2020 - April 2021

- Developed a Strain Characterization Apparatus capable of high precision ($\pm 0.5\%$) and high accuracy ($\pm 5\%$) to determine Young's modulus values of various materials under homogeneous stress conditions through lab tests.
- Partnered with MesoMat, to design device to test proprietary strain sensors, which earned 2nd place of all capstone projects.
- All in one embedded device with data collection, display, parsing and cloud database saving.

EDUCATION

Bachelor of Engineering, Software

McMaster University • Hamilton, ON • 2021

SKILLS

Languages: C++, C, Java, Python, Bash, JavaScript, HTML, CSS

Software: GitHub, Git, Jenkin, Jira, Confluence, Android Studio, Visual Studio, Perforce

Technical Skills: DisplayPort, HDMI, AVC, HEVC, Video Encoding, Multimedia, ML/AI, SQL Databases